cluster, all 100 nodes should have the same `cluster.name` property. The default cluster name for an out-of-the-box server is `elasticsearch`, but it is a best practice to configure a unique name for the cluster.

The cluster can be ramped up by scaling up (vertical scaling) or scaling out (horizontal scaling). When additional nodes are booted up, they can join the same cluster as existing nodes as long as the `cluster.name` property is the same. Thus, a group of nodes can form a multi-node cluster, as shown in figure 3.15.

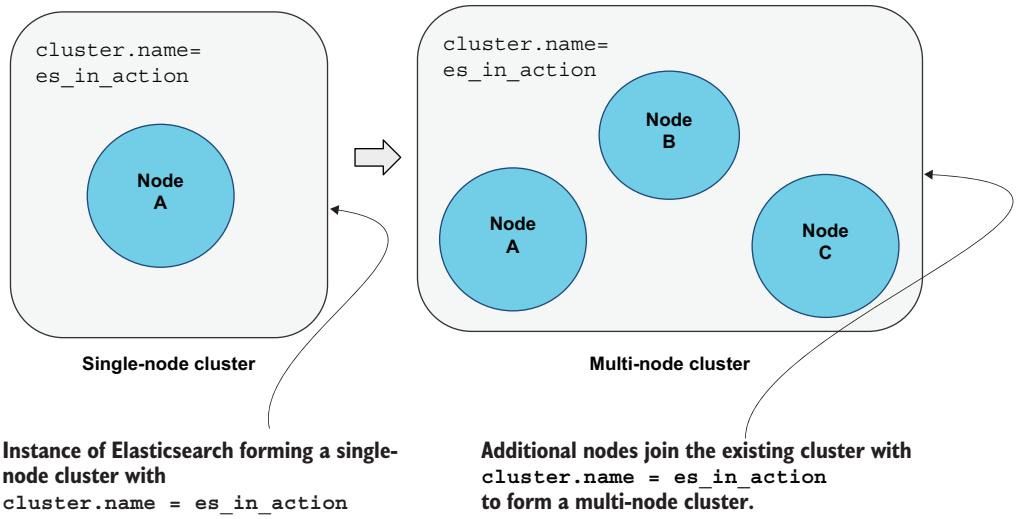


Figure 3.15 Cluster formation from a single node to a multiple-node cluster

Adding more nodes to the cluster not only creates redundancy and makes the system fault tolerant but also brings huge performance benefits. When we add more nodes, we create more room for replicas. Remember that the number of shards cannot be changed when the index is in operation. So, what's the benefit of shards adding nodes? Usually, data is reindexed from an existing index into a new index. This new index is configured with a new number of shards, taking the additional new nodes into consideration.

Manage node disk space judiciously

To improve read performance, we can add replicas, but with them come greater memory and disk space requirements. While it is not unusual to create clusters containing terabytes or even petabytes when working with Elasticsearch, we must consider our data sizing requirements.